University of Florida, Department of Mathematics

Analysis PhD Exam, January 1997

I.

State the following theorems.

1. The monotone convergence theorem for nets in L^p.
2. The Hahn decomposition theorem.
3. The Egorov theorem.
4. The Vitali convergence theorem.
5. The Radon–Nikodym theorem.
6. The Fubini theorem.
7. The representation of the dual of L^p.
8. The Lebesgue dominated convergence theorem.

II

Prove one of the theorems 6 or 7.

III.

Solve 5 of the following problems.

In what follows, (X, Σ, μ) is a measure space.

1. Let (f_n) be a sequence of real-valued measurable functions on X, and let f be a real measurable function on X. Prove that (f_n) converges to f in μ-measure, $f_n \xrightarrow{\mu} f$, iff every subsequence (f_{n_i}) contains a further subsequence $(f_{n_{i_j}})$ converging to f, μ-a.e.

2. Let (f_n) be a sequence of positive, μ-integrable functions such that $\sum_n \int f_n \, d\mu < \infty$. For each $x \in X$, denote
$$L(x) = \sup\{n : f_n(x) > 1\}.$$
Prove that $L(x) < \infty$, μ-a.e.

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a Lebesgue integrable function. Prove that for every $\varepsilon > 0$, there is a bounded interval I such that $\left|\int_E f \, dx\right| < \varepsilon$ for every measurable set E with $E \cap I = \varnothing$.

4. Let F be a Banach space and $\mathcal{F} \subseteq \Sigma$ a sub-σ-algebra. Prove the existence of the conditional expectation $E(f \mid \mathcal{F})$ for every μ-integrable function $f : X \to F$.

5. Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be two real measure spaces. Prove the existence of the product measure $\mu \times \nu$ and that $|\mu \times \nu| = |\mu| \times |\nu|$.

6. Let μ be the Lebesgue measure on $\mathbb{R}$, and let f be a Lebesgue integrable function on $\mathbb{R}$ such that $\int_0^x f \, d\mu = 0$ for every $x \in \mathbb{R}$ (if $x < 0$, $\int_0^x = -\int_x^0$). What can you say about f?

7. Let μ be the Lebesgue measure on $\mathbb{R}$, and let $f_n = -\varphi_{(n,\infty)}$, $n = 1, 2, \ldots$. Show that (f_n) is increasing but that the conclusion of the monotone convergence theorem is not true. Explain why.

8. Let μ be the Lebesgue measure on $(0, 1)$. Each point $x \in (0, 1]$ has a binary expansion

$$x = \sum_{i=1}^{\infty} \frac{x_i}{2^i},$$

where $x_i = 0$ or 1. (The expansion is unique except on a countable set, which we can throw out of the space.)

(i) Define $T_i(x) = x_i$. Sketch the graphs of T_1, T_2, T_3.

(ii) Let $\mathcal{F}_k = \sigma(T_1, T_2, \ldots, T_k)$. Describe $\mathcal{F}_k$.

(iii) Let $f \in L^2(\mu)$. Find a function g_1 which is $\mathcal{F}_1$-measurable and satisfies

$$\int_A g_1 \, d\mu = \int_A f \, d\mu \quad \text{for } A \in \mathcal{F}_1.$$

That is, find $g_1 = E(f \mid \mathcal{F}_1)$. Find $g_2 = E(f \mid \mathcal{F}_2)$.